GG-S-86D
5 November 1985
SUPERSEDING
GG-S-86C
25 September 1973

FEDERAL SPECIFICATION
SAFELIGHT, DARKROOM, PHOTOGRAPHIC

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for use of all Federal agencies.

1. SCOPE

1.1 Scope. This specification covers safelights for use in photographic and x-ray darkrooms. Filters and lamps for use in the safelights are not included.

1.2 Classification. Safelights shall be the following types, styles, and sizes, as specified (see 6.2):

Type I - Safelight only, with no provision for white light.
Type II - Combination safelight and white viewing light (size B only).

Style 1 - For wall mounting.
Style 2 - For bench mounting.
Style 3 - For ceiling mounting.

Size A - For 5 by 7 inch (12.7 by 17.8cm) filers.
Size B - For 8 by 10 inch (20.3 by 25.4cm) filters.
Size C - For 10 by 12 inch (25.4 by 30.5cm) filters.
Size D - For 5.5 inch (14cm) diameter filters.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

W-C-596 - Connector, Cable, Outlet, Electrical, Specific Purpose, Cable Connecting, General Grade, Locking, 2 Pole, 2 Wire, 15 Amperes, 125 Volts, 50/60 Hertz.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Electronics Research and Development Command, ATTN: DELCS-PE, Fort Monmouth, New Jersey 07703, by using the self-addressed Standardization Document Improvement Proposal (DD FOrm 1426) appearing at the end of this document or by letter.

FSC 6740

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W-L-101 - Lamp, Incandescent, (Electric, Large Tungsten Filament).
W-L-142 - Lampholder, Adapter, And Shadeholder, Medium-Screw-Shell, 125, 250, And 600 Volts.

MILITARY

MIL-P-116 - Preservation, Methods Of.
MIL-P-11268 - Parts, Materials, And Processes Used In Electronic Equipment.
MIL-E-55585 - Electronics Equipment And Parts, Packaging Of.

STANDARDS

FEDERAL

FED-STD-123 - Marking For Shipment (Civil Agencies).

MILITARY

MIL-STD-105 - Sampling Procedures And Tables For Inspection By Attributes.
MIL-STD-130 - Identification Marking For U.S. Military Property.
MIL-STD-252 - Wired Equipment, Classification Of Visual And Mechanical Defects.
MIL-STD-454 - Standard General Requirements For Electronic Equipment.
MIL-STD-726 - Packaging Requirements Code.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/NFPA 70 - National Electrical Code.

(Application for copies should be addressed to the American National Standards Institute Inc., 1430 Broadway, New York, NY. 10018.)

UNDERWRITERS' LABORATORIES, INC. (UL)

UL 57 - Electric Lighting Fixtures.

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

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(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Description. The safelight shall consist of a lamp house, with lamp socket, attachment cord, switch and attachment plug as specified herein.

3.2 First article. When specified (see 6.2) the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.3 and 6.3).

3.3 Parts, materials and processes; general. In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection requirement therein, shall apply.

3.3.1 Recycled, virgin and reclaimed materials. It is required that, to the maximum extent possible, recovered or reclaimed materials, in lieu of virgin materials, shall be used without jeopardizing the intended use of the manufactured item.

3.4 Design.

3.4.1 Lamp house. The lamp house shall be steel not less than 0.025 inch (.06cm) thick, or of other suitable material and thickness adequate for providing an equivalent strength and rigidity required for lasting alignment of parts and complete freedom from stray light leaks. All seams shall be overlapped or interlocked throughout and so joined by welding, spot welding, or other means suitable to provide permanent rigidity and to prevent any possibility of light leaks. Electrical outlets, cords, switches, mounting devices, and other parts of, or attachments to the lamp house shall be so designed and installed as to prevent any stray light leaks.

3.4.2 Mounting facilities and devices.

3.4.2.1 Style 1 (wall mounting). Size A and size B safelights shall be provided with eyelets or keyhole slots to allow the safelight to be mounted on a wall. Size C safelights shall be provided with a metal bracket which has eyelets or keyhole slots for wall mounting, and which will allow rotation of the safelight about its long axis through an angle of not less than 90°. The bracket shall be provided with a means for retaining the lamp house at any set degree of rotation. Size C safelights shall also be provided with a metal ring at each of their four top corners to permit ceiling mounting. Size D safelights shall be provided with a bracket and base plate to allow mounting on a wall, with adjustment of the lamp house to any desired angle. The mounting facility shall be provided with a means for retaining the lamp house at the set degree of tilt.

3.4.2.2 Style 2 (bench mounting). Size A, B, and C safelights shall be designed to stand firmly on a shelf, bench, table top, or other flat, horizontal surface without the need for special mounting. Size D safelight shall be provided with a bracket and base plate to allow being mounted to a flat, horizontal surface, with provision for adjusting the lamp house to any desired angle. The mounting facility shall be provided with a means for retaining the lamp house at the set degree of tilt.

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3.4.2.3 Style 3 (ceiling mounting). Size A, B, and C safelights shall be provided with a metal ring at each of their four corners to allow attachment of chains for suspension of the safelights from the ceiling. Four chains, each not less than 44 inches (111.8cm) long, shall be provided for this purpose. In addition, when specified, other ceiling-mounting devices shall be furnished (see 6.2). Size D safelights shall be provided with a medium-screw base rigidly attached to the lamp house, to allow the safelight to be screwed into a standard medium-screw lamp socket.

3.4.3 Filter holder. Safelights of the types, styles, and sizes specified herein shall be provided with suitable means for receiving and holding filters of appropriate sizes as specified in 1.2. The allowable tolerances for the length and width dimensions of rectangular filters are +1/32 or -1/16 inch (+.08 or -.16cm). Thickness shall be 0.160 ±0.02 inch (.41 ±.05cm). Size D safelights shall accommodate circular filters 5.5 (+1/32 or -1/16) inch ,14 (+.08 or -.16cm) in diameter, 0.24 ± 0.02 inches (0.61 ±.05cm) thick. Dimensions of filter holders shall be such as to furnish adequate, but not excessive, clearances so that there shall be no possibility of damage to the filters, lamp houses, or filter holders from strains or other mechanical faults due to forced fits. Filter holders and retaining devices shall be so constructed that there is no leakage of "white" light when the lamp house-filter-filter holder assembly is in use. It shall be possible to remove and replace filters without the use of any tools other than a common flat-blade screw-driver. In no case shall the filter holder be secured by screws of the "self-tapping" type or other means whereby repeated use may result in loosened or otherwise ineffective holding.

3.4.4 Electrical fittings. All electrical fittings and wiring shall be in accordance with ANSI/NFPA 70 as specified herein.

3.4.4.1 Sockets. Safelights shall be provided with medium-screw-base socket(s) to accommodate the recommended lamp(s). Sockets shall be rigidly, but removably, attached to the lamp house and shall conform to W-L-142.

3.4.4.2 Attachment cord. With the exception of style 3, size D safelights, and specially specified ceiling mounts, a flexible rubber or plastic-covered cord of not less than 6 feet (183cm) in length shall be provided as an integral part of the safelight. The cord shall terminate in a molded-rubber attachment plug conforming to W-C-596. The cord shall be the 3-conductor type, having one conductor color-coded or otherwise marked permanently attached to the metal part of the safelight and the other end terminated at the grounding prong of a three-prong plug, as specified (see 6.2). The insulation between the conductors shall conform to W-C-596.

3.4.4.3 Switches. Switches shall be as specified (see 6.2). Switches provided for type II safelights shall be so arranged by orientation or electrical interconnection as to prevent inadvertently turning on the white light when a safe light is desired. In no case shall so-called "sequential" switches be used.

3.4.5 Viewing light. Type II, styles 1 and 2, size B safelights shall be provided with a white-light viewing panel which shall be designed to accommodate an opal glass slide 10 ±0.06 inches (25.4 ±.15cm) long, by 4.75 ±0.06 inches (12.07 ±.15cm) wide by 0.10 ±0.02 inch (.25 ±.05cm) thick. Glass slide holder dimensions and ease of replacement of the glass shall be as specified in 3.4.3.

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3.5 Environmental requirement.

3.5.1 Temperature rise. The design, materials, and construction of the safelights shall be such as to permit their continuous operation with no more than a 40°F (22°C) rise above ambient temperature when tested in accordance with 4.6.2.

3.6 Incandescent lamp. Each safelight shall be furnished with an information sheet or other means of notification of the maximum wattage of the lamp(s) and working distance to be used with the safelight. The lamps shall be rated for 120-volt, 60-Hertz operation, and shall conform to W-L-101.

3.7 Fire and casualty hazards. The contractor shall submit proof to the contracting officer that the safelights supplied conform to UL 57, Electrical Lighting Fixtures. The labels or listing with re-examination of the Underwriters' Laboratories Inc., may be accepted as evidence that the safelights conform to this requirement.

3.8 Exterior Finish. Unless otherwise specified (see 6.2) the exterior finish of the safelight shall be smooth baked enamel and the color shall be the contractor's option.

3.9 Marking. Marking shall conform to MIL-STD-130 for military requirements, and FED-STD-123 for civil agencies.

3.10 Workmanship. The safelight shall be manufactured and assembled in accordance with MIL-STD-454, requirment 9, with particular attention to the following: welding; freedom from blemishes, defects, burrs, sharp edges; rigidity of seams and electrical connections.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Inspections covered by subsidiary documents (see 4.4).
c. Quality conformance inspections (see 4.5).
d. Packaging inspection (see 4.8).

4.3 First article.

4.3.1 First article units. When specified, the contractor shall furnish first article units of the complete safelight.

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4.3.2 First article inspection. The first article inspection shall consist of the inspections specified in subsidiary documents covering the items in 4.4, and the inspections specified for group A, group B, and group C (see tables I, II, III respectively). The inspection shall be performed in the following order: 4.4, group A and group B for all units, and group C as specified in table III. Failure of any inspection shall be cause for rejection.

4.4 Inspections covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification, and the inspection requirement specified in the contract or purchase order:

Item	Where required
Sockets	3.4.4.1
Connector	3.4.4.2
Marking	3.9

4.5 Quality conformance inspection. The contractor shall perform the inspections specified in 4.4 and 4.5.1 through 4.5.3.2.

4.5.1 Group A inspection. Each unit shall be inspected for conformance to the inspections specified in table I. Discrete lots shall be formed from units that pass inspection. Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be subjected to sampling inspection, utilizing the procedures of MIL-STD-105, using the general inspection levels, and acceptable quality levels (AQL) specified in table I.

TABLE I. Group A inspection.

Inspection	Requirement paragraph	Test method paragraph	AQL (percent defective) Major	Minor
Visual and mechanical	3.10	4.7	1.0	4.0

4.5.2 Group B inspection. Group B inspection shall normally be performed on inspection lots that have passed group A inspection and on samples that have been subjected to and met the group A inspection. This inspection shall be as specified in table II.

TABLE II. Group B inspection.

Inspection	Requirement paragraph	Test method paragraph
Light-tightness	3.4.3	4.6.1

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4.5.2.1 Group B sampling plan. Group B sampling plan shall be in accordance with MIL-STD-105, inspection level S-4, AQL shall be 6.5 percent defective.

4.5.3 Group C inspection. Group C inspection shall be performed on units that have passed group A and B inspection. The inspection shall be as specified in table III.

TABLE III. Group C inspection;

Inspection	Requirement paragraph	Test method paragraph
Temperature rise	3.5.1	4.6.2

4.5.3.1 Sampling for group C inspection. One sample unit shall be selected at random for the inspection in table III from each 200 or fraction thereof produced. The first sample selected shall be at the start of the contract from the quality conformance inspection production lot.

4.5.3.2 Group C failure. Actions required relative to group C failure shall be as specified (see 6.2).

4.6 Methods of inspection.

4.6.1 Light tightness. The safelight shall be tested as follows: A lamp of the maximum recommended wattage shall be installed. An opaque slide dimensioned to the filter size shall be installed in the filter holder. The safelight shall be mounted in its normal operating position in a totally darkened room. Sheets of unexposed Eastman Kodak Co. Tri-X film or equal, not less than 2.25 by 3.25 inches (5.72 by 8.26cm) shall be placed in positions where film would normally be handled. The safelight shall be turned on. At the end of 30 minutes, the film shall be processed and examined for evidence of light fog. Nonconformance to 3.4.1 shall constitute a failure of this test.

4.6.2 Temperature rise. The safelight shall be operated under conditions of normal use in an ambient temperature of 90°F (32°C), with a lamp of maximum recommended wattage and at rated voltage. The safelight shall have a green filter of suitable size installed in the filter holder. The safelight shall be operated for not less than one hour, or until thermal equilibrium is established. The temperature of the hottest part of the filter shall be determined by means of a thermocouple attached to the filter by tape or other suitable material. The ambient temperature and the thermocouple temperature reading shall be recorded. Nonconformance to 3.5.1 shall constitute failure of this test.

4.6.3 Fire and casualty hazards. The safelight shall be inspected for compliance with ANSI/NFPA 70 and UL 57. Nonconformance to 3.4.4 and 3.7 shall constitute failure of this test.

4.7 Visual and mechanical inspection. The safelight shall be examined for the defects listed in MIL-STD-252. Nonconformance to 3.10 shall constitute failure of this test.

4.8 Packaging inspection. Packaging inspection requirements specified herein are classified as follows:

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a. First article inspection of packaging.
b. Quality conformance inspection of packaging.

4.8.1 First article inspection of packaging. Unless otherwise specified in the contract, first article inspection of packaging shall be in accordance with the unit pack design validation requirements of MIL-P-116.

4.8.2 Quality conformance inspection of packaging.

4.8.2.1 Materials inspection. All materials to be used in packaging shall be inspected in accordance with the applicable material specification.

4.8.2.2 Preservation inspection. Inspection of preservation and interior markings shall be in accordance with group A and B quality conformance inspection requirements of MIL-P-116. Lot formation and sampling procedures shall be as specified therein.

4.8.2.3 Packing inspection. Inspection of packing and the marking for shipment and storage shall consist of the examinations specified in table IV, "Packing inspection provisions." Lot formation shall consist of all packs made of the same materials during an identifiable period and submitted at one time for acceptance. Sampling procedures shall be in accordance with MIL-STD-105, using a single sampling plan and acceptable quality level of 4.0 percent defective.

TABLE IV. Packing inspection provisions.

No.	Characteristic	Method of inspection
101	Intermediate container not as specified.	Visual
102	Improper closure of intermediate container.	Visual
103	Shipping containers not in accordance with specification.	Visual
104	Excessive cube.	Visual
105	Improper blocking and bracing.	Visual
106	Closure not in accordance with specification.	Visual
107	Weight and size exceed container limitations.	Weight & measure
108	Strapping not in accordance with specification, incorrectly applied or omitted.	Visual
109	Marking omitted, incorrect, or illegible.	Visual

5. PACKAGING

5.1 Preservation. Preservation shall be as specified in MIL-STD-726, coded as follows:

5.1.1 Level A. 1ϕ-1-1-ϕϕ-00-NS-X-ED-ϕ-00-A.

5.1.2 Level B. 1ϕ-1-1-ϕϕ-00-NS-X-ED-ϕ-00-B.

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5.2 Packing and marking. Packing and marking shall be in accordance with MIL-E-55585.

6. NOTES

6.1 Intended use. Safelights are used in photographic and X-ray darkrooms to assure safe handling of light sensitive camera film or material.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type, style, and size safelight required (see 1.2).
c. When a first article is required for inspection and approval (see 3.2).
d. When ceiling mounting devices shall be other than specified (see 3.4.2.3).
e. Type attachment plug required (see 3.4.4.2).
f. Type of switch, if any, required (see 3.4.4.3).
g. Type of finish and color (see 3.8).
h. Group C failure action (see 4.5.3.2).
i. Level A or B preservation and packing (see section 5).
j. When first article inspection rough handling tests are not required.
k. When first article packaging inspection test reports require acquisition activity approval prior to production unit packing.

6.4 Level B preservation. When level B preservation is specified, this level of protection will only be used under known favorable conditions during transportation, storage and handling.

6.5 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the change.

Custodians:
Army - ER
Navy - AS
Air Force - 99

Preparing activity:
Army - ER

(Project 6740-0139)

Review activities:
Army - ME, MD

User activity:
Navy - MC

Civil Agencies Coordinating Activity:
GSA - FSS

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
GG-S-F6D	SAFELIGHT, DARKROOM, PHOTOGRAPIC

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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